23/11/2021

README for information on raw data files relating to manuscript:

"*Identification of a stereotypic molecular arrangement of endogenous glycine receptors at spinal cord synapses*" S.A. Maynard, P. Rostaing, N. Schaefer, O. Gemin, A. Candat, A. Dumoulin, C. Villmann, A. Triller, C.G. Specht. eLife 2021.

Animal IDs for Imaging:

>2 months WT
AQ847 EosGlyRb KI/wt mRFP-GPHN KI/KI ♀
AQ848 EosGlyRb KI/KI mRFP-GPHN KI/KI ♀
AQ858 EosGlyRb wt/wt mRFP-GPHN KI/KI ♀
AQ859 EosGlyRb KI/KI mRFP-GPHN KI/wt ♀
>10 months WT
BP057 EosGlyRb KI/KI mRFP-GPHN KI/KI ♀
BP001 EosGlyRb KI/KI mRFP-GPHN KI/wt ♀
BP056 EosGlyRb KI/wt mRFP-GPHN KI/KI ♀
>2 months *Oscillator*
F2-1 EosGlyRb KI/KI oscillator KI/wt ♀
F2-5 EosGlyRb KI/KI oscillator KI/wt ♀
F2-2 EosGlyRb KI/KI oscillator wt/wt ♀ (WT littermate)
F2-3 EosGlyRb KI/KI oscillator wt/wt ♀ (WT littermate)

Confocal Image Data:

>180521 double KI ICC 7a quantified. lif
>Contains all images for decay, intensity measurements and cross-section quantification of 2 month WT mice
>Figure 1

Decay:
    AQ848
Intensity measurements:
    AQ847
    AQ848
    AQ858
    AQ859
Quantification cross-section:
    AQ847
    AQ848

>190117 double KI ICC b3 quantified.lif
>Contains all images for decay, intensity measurements and cross-section quantification of 10 month WT mice
>Figure 1 and Figure 1-figure supplement file 3, 4

Intensity measurements:
- BP056
- BP057

Quantification cross-section:
- BP056
- BP057

>210119 osci KI ICC geph7a VGAT.lif
>Contains images for cross-section quantification of 2 month oscillator vs WT littermate mice
>Figure 4 and Figure 4-figure supplement file 1

Quantification cross-section:
- F2-1
- F2-2

>210120 osci KI ICC geph7a VGAT.lif
>Contains images for cross-section quantification of 2 month oscillator vs WT littermate mice
>Figure 4 and Figure 4-figure supplement file 1

Quantification cross-section:
- F2-3
- F2-5

PALM Image Data:

>PALM IDs:

180919: AQ859
- 001-002: Dorsal

180920: AQ859
- 003-006: Ventral
- 007-009: Dorsal
- 010-014: Ventral

180922: AQ859
- 015-019: Dorsal

181130: AQ848
- 001-009: Dorsal

181201: AQ848
- 010-017: Ventral
- 018-021: Dorsal

181202: AQ848

022-027: Ventral
190312: BP057
001-005: Dorsal
006-010: Ventral
190313: BP057
011-015: Dorsal
016-020: Ventral
190320: BP001
001-005: Ventral
006-009: Dorsal
190321: BP001
010-015: Dorsal
016-020: Ventral
210120:
001-006: Ventral F2-1
007-012: Dorsal F2-1
013-015: Ventral F2-2
210122:
016-018: Ventral F2-2
019-024: Dorsal F2-2
210128:
038-043: Ventral F2-3
044-049: Dorsal F2-3
210129:
050-055: Ventral F2-5
056-061: Dorsal F2-5

>PALM images

>File names: _MTT_without_tracking_results.mat
>Images used for reconstructed image analysis for ICQ and occupancy analysis
>Figure 2, figure 5 and Figure 2-figure supplement file 1

MATLAB files contain columns of all the detections.
Row titles:
frame number
X position
Y position
alpha (intensity under the curve)
sigma (radius of Gaussian profile)

>PALM detection & synapse area data

>File names: -roi1-counts.txt
>Images used for synapse area, number of synapses and detections
>Figure 2, figure 3, figure 5 and Figure 2-figure supplement file 2, Figure 2-figure supplement file 4, Figure 3-figure supplement file 1, Figure 5-figure supplement file 1

Text files (-roi1-counts.txt) contain columns of all the synapses.

Column titles:

Cluster number
X position
Y position
Cluster area (#10x10np pixels)
Bursts
#Detections per cluster
#Detections/Burst

>PALM molecule conversion

>File names: -roi1-countsbursts.txt
>Images used for molecule conversion calculation
>Figure 2, figure 3, figure 5 and Figure 2-figure supplement file 3, Figure 2-figure supplement file 4, Figure 5-figure supplement file 2

Text files (-roi1-countsbursts.txt) contain columns of burst counts.

Column titles:

#Bursts
Counts
Mean # of detections

SRRF Images:

SRRF reconstructed images
>File names: _fr100-6000-SRRF.tif

>SRRF IDs:
Same as PALM AQ859, AQ848, BP057 and BP001

>Figure 2 and Figure 2-figure supplement file 1

Electron Microscopy Images:

>EM Images:

190312

Dorsal 1 grid C position G10 (D1 GC G10): Synapses 3, 4, 5
Ventral 1 grid B position G5 (V1 GB G5): Synapses 1, 2, 4, 5, 7, 8, 9 10, 11, 13

190313

Dorsal 2 grid B position E9 (D2 GB E9): Synapses 1, 2, 3, 4, 5, 6, 7
Dorsal 2 grid B position G9 (D2 GB G9): Synapses 1, 2, 3
Ventral 2 grid C position E3 (V2 GC E3): Synapses 4, 7, 8, 9, 10, 13, 14, 15, 16, 17, 18, 19, 22

>Images of serial sections used to measure area of postsynaptic domain of

synapses imaged with PALM and SRRF.

>Figure 3 and Figure 3-figure supplement file 1

Electrophysiology:

>Cell IDs:

Wt
Exp 1 (Bl6_wt_exp1.opju)
Contains data from
neuron 1
neuron2
Exp 2 (Bl6_wt_exp2.opju)
Neuron 4
Neuron 5
Neuron 6
Neuron 7
Neuron 8
Exp 3 (Bl6_wt_exp3.opju)
Neuron 9
Neuron 10
Neuron 11

Eos
Exp1 (eos_exp1.opju)
Neuron 1
Neuron 2
Exp2 (eos_exp2.opju)
Neuron 3
Neuron 4
Neuron 5
Neuron 6
Neuron 7
Exp 3 (eos_exp3.opju)
Neuron 8
Neuron 9
Neuron 10
Neuron 11

>Electrophysiology data:

>All data files were from HEKA exported into origin, the origin files contain xy date (current versus time).
>The Originlab, Origin 2020 was used to setup the data files.
>Figure 1-figure supplement file 2